

Data Analysis Programming

Semester 2026-I

Workshop No. 3 — Silver-to-Gold Pipeline, Governance KPIs, and Storytelling Definition

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Welcome to the third workshop of the *Data Analysis Programming* course!

In **Workshop #1**, you defined your project, validated your data sources, and established the extraction strategy. In **Workshop #2**, you built the technical infrastructure, implemented the Bronze ingestion DAG and the FileSensor-triggered Silver processing DAG, and produced your first pipeline outputs. Now, you will iterate on your previous work, improve your pipeline based on real experimentation findings, and implement the **Gold layer**: the analytical heart of the Medallion Architecture.

This workshop focuses on three interconnected goals: (1) improving the Bronze and Silver pipelines based on lessons learned, (2) implementing the Silver-to-Gold transformation using PySpark to load and aggregate multiple Parquet files, and (3) defining the content, KPIs, and storytelling narrative for both Gold-layer dashboards.

General Workshop Definition: The Gold layer represents the analytical and decision-support tier of the data pipeline. In this workshop, teams will use PySpark to read and consolidate multiple Silver Parquet files — simulating a cloud storage folder read pattern — and produce two structured summary datasets: one for data governance and one for functional user storytelling. Both summaries will be persisted in `datalake_gold/` as Parquet files and will serve as the sole data source for the Plotly Dash dashboards built in Workshop #4.

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Workshop Scope and Objectives

- **Pipeline Improvements:** Revisit and improve the Bronze and Silver pipelines from Workshop #2 based on real experimentation findings, data quality observations, and NLP processing results.
- **Silver-to-Gold DAG with PySpark:** Implement a scheduled Airflow DAG that reads all accumulated Parquet files from `datalake_silver/`, performs aggregations and NLP analysis using PySpark, and persists two summary datasets in `datalake_gold/`.
- **Governance Summary Definition and Implementation:** Define and compute KPIs for data governance, reflecting data quality, volume, completeness, and structural health of the ingested data.
- **Storytelling Summary Definition and Implementation:** Define and compute the analytical aggregations that will support the functional user dashboard: sentiment trends, keyword distributions, temporal patterns, and topic insights.
- **Data Quality Findings Documentation:** Produce a structured report of data quality findings observed in the Silver layer as justification for governance KPI selection.

Pipeline Improvement and Gold Layer Methodology

1. Pipeline Improvements Based on Experimentation:

- Review all Bronze JSON files and Silver Parquet files produced in Workshop #2 and document observed data quality issues not anticipated in the initial pipeline design
- Improve the preprocessing pipeline (Silver DAG) to address newly identified issues: additional null fields, text encoding problems, inconsistent date formats, new outlier patterns, or schema deviations
- Refine NLP text preprocessing steps based on actual text content: adjust stop word lists, add domain-specific noise removal (URLs, hashtags, mentions, HTML tags), or improve tokenization
- If the Bronze ingestion DAG had reliability issues, apply improvements: better error handling, retry logic, or more robust scraping selectors
- Document all improvements with a clear before/after comparison, explaining what was observed, what was changed, and why

2. Silver-to-Gold DAG with PySpark:

- Implement an Airflow DAG scheduled to run periodically (e.g., `@weekly` or a defined cron expression) that orchestrates the Silver-to-Gold transformation

- Use **PySpark** within the DAG tasks to read all Parquet files from `datalake_silver/` as a single distributed DataFrame, simulating a cloud storage folder read pattern (`spark.read.parquet("datalake_silver/")`)
- Apply distributed aggregations and transformations using PySpark DataFrame API or Spark SQL
- Produce two output datasets and persist each as a Parquet file in `datalake_gold/` with a timestamp-based filename
- Configure PySpark to run in local mode inside the Docker environment (e.g., `SparkSession.builder.master("local[*]")`)
- Document the Spark session configuration, memory settings, and any Docker adjustments required to run PySpark within Airflow

3. Governance Summary — KPI Definition and Implementation:

- Define a minimum set of **data governance KPIs** grounded in your actual Silver data quality observations. Suggested KPIs include:
 - *Null rate per field*: Percentage of null or empty values for each column across all ingested records
 - *Volume metrics*: Total records ingested per source, per time period, and cumulative
 - *Duplicate rate*: Percentage of duplicate records detected before and after deduplication
 - *Schema compliance rate*: Percentage of records conforming to the expected Silver schema
 - *Outlier rate per numeric field*: Percentage of records flagged as outliers per numeric column
 - *Text length statistics*: Mean, median, min, and max text length per source and time period
 - *Language distribution*: If multilingual data exists, percentage breakdown by detected language
 - *Ingestion frequency compliance*: Actual vs. expected ingestion runs per time period
- Justify each selected KPI with reference to the actual data quality findings from your Silver pipeline
- Implement all KPI computations using PySpark and persist the governance summary as a structured Parquet file in `datalake_gold/governance_YYYYMMDD_HHMMSS.parquet`

4. Storytelling Summary — Analytical Aggregations Definition and Implementation:

- Define the analytical aggregations that will populate the functional user dashboard, aligned with the user stories defined in Workshop #1. Suggested aggregations include:

- *Sentiment distribution*: Count and percentage of positive, negative, and neutral records overall and per time period
- *Sentiment trend over time*: Average sentiment score per day/week to reveal temporal opinion shifts
- *Top keywords and n-grams*: Most frequent terms and bigrams extracted from the preprocessed text corpus
- *Keyword sentiment association*: Average sentiment score associated with the most frequent keywords
- *Source comparison*: Sentiment distribution and volume breakdown by data source (API vs. scraping)
- *Volume trends*: Number of records per time period to identify topic activity peaks
- *Topic or category breakdown*: If applicable, sentiment and volume distribution across subtopics or categories
- Justify each aggregation with reference to a specific user story from Workshop #1
- Implement all aggregations using PySpark and persist the storytelling summary as `datalake_gold/storytelling_YYYYMMDD_HHMMSS.parquet`
- Define the dashboard narrative: what story does the data tell? What key insight should the functional user walk away with after viewing the dashboard?

5. Data Quality Findings Report:

- Produce a structured section in the workshop report documenting all data quality findings observed in the Silver layer
- For each finding, describe: the field or column affected, the nature of the issue (nulls, outliers, encoding, schema deviation), the frequency or severity, and the handling strategy applied
- Use this findings report as the explicit justification for the governance KPIs selected
- Include descriptive statistics tables and visualizations (from a Jupyter Notebook) to support the findings

Deliverables and Documentation

1. Pipeline Improvements and Gold Layer Report (PDF):

- *Executive Summary*: Overview of improvements made and Gold layer implementation approach
- *Pipeline Improvements*: Before/after comparison of Bronze and Silver pipeline changes with justification

- *Data Quality Findings*: Structured report of observed Silver data quality issues with statistics and visualizations
- *Governance KPI Definition*: Complete description of each KPI, its computation logic, and justification based on data findings
- *Storytelling Aggregations*: Complete description of each analytical aggregation, its mapping to a user story, and the dashboard narrative
- *PySpark Implementation*: Description of the Spark session setup, DAG structure, and key transformation steps
- *Results Evidence*: Screenshots of Gold Parquet file contents for both governance and storytelling summaries

2. Implementation Code:

- Updated Bronze and Silver DAG files with all improvements applied
- Gold DAG file (`airflow/dags/gold_processing_dag.py`) with PySpark transformations
- Any supporting PySpark transformation modules or utility functions
- Updated `docker-compose.yml` if PySpark integration required additional configuration

3. Exploratory Notebooks:

- A Jupyter Notebook in `notebooks/` demonstrating data quality findings: descriptive statistics, null rates, outlier analysis, and text length distributions using Pandas
- A Jupyter Notebook demonstrating how to read the Gold Parquet files and preview the governance and storytelling summaries

4. Repository Update:

- Add a `workshop_3/` folder containing the report PDF and any additional documentation
- Update the repository `README.md` to reflect the current three-layer pipeline status and how to trigger the Gold DAG
- Include sample Gold Parquet files in the repository for both governance and storytelling datasets

Submission Requirements

- Submit your complete Pipeline Improvements and Gold Layer Report as a single PDF through the designated course platform
- Include your GitHub repository link with all updated code, Gold DAG implementation, and Gold Parquet output samples
- All documentation must be in **English** and follow professional technical writing standards
- Ensure proper citations for PySpark documentation, NLP techniques, and any referenced data quality frameworks
- The Gold DAG must be functional and produce both governance and storytelling Parquet files — include evidence of successful runs in the report

Important Guidelines and Considerations

- **PySpark in Local Mode:** PySpark does not require a cluster to be useful. Running in `local[*]` mode inside Docker is perfectly valid for this project and mirrors the pattern of reading from a cloud storage folder. Focus on learning the DataFrame API and Spark SQL, not cluster management.
- **Multiple Parquet Files as a Folder:** The key PySpark pattern here is `spark.read.parquet("datalake_silver/")` — this reads all Parquet files in the folder as a single DataFrame, exactly as you would read from an S3 bucket or Azure Data Lake in production. Ensure your Silver Parquet files have consistent schemas across runs.
- **KPIs Grounded in Data:** Do not define governance KPIs theoretically. Every KPI must be justified by an actual data quality observation from your Silver pipeline. If your data has no duplicates, document that as a finding and explain why a duplicate rate KPI is still included as a preventive measure.
- **Storytelling Narrative:** The storytelling summary is not just a collection of statistics — it must tell a coherent story about public opinion on your topic. Define the narrative first (what you want the user to understand), then define the aggregations that support it.
- **Experimentation Mindset:** This workshop explicitly values the process of experimenting, observing unexpected results, and improving. Document your experimentation process, including things that did not work and why, as this demonstrates engineering maturity.
- **Gold as the Single Source of Truth for Dashboards:** The Plotly Dash applications in Workshop #4 must read exclusively from the Gold Parquet files. Design your Gold summaries with the dashboard layout in mind — each aggregation should map directly to a chart or KPI card in the dashboard.

Good luck, and embrace the analytical depth of this workshop — the Gold layer is where raw data transforms into genuine insight, and where your project begins to answer the questions your functional user actually cares about!